

SMART SECURITY SYSTEM

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SMART MUSEUM ARTIFACT SECURITY SYSTEM

An Intelligent ASM-Based Security System
for Artifact Protection and
Environmental Safety



MOTION DETECTION
In Safe Zone



**ACCESS CONTROL
& VERIFICATION**



**FIRE / SMOKE
DETECTION**



**GAS LEAKAGE
DETECTION**



ALARM & LOCKDOWN
System Response



EMERGENCY RESPONSE
Emergency Team Dispatched



DETECT



VERIFY



ALERT



LOCKDOWN



EMERGENCY RESPONSE



PROJECT TYPE
Final Project
Verilog FPGA



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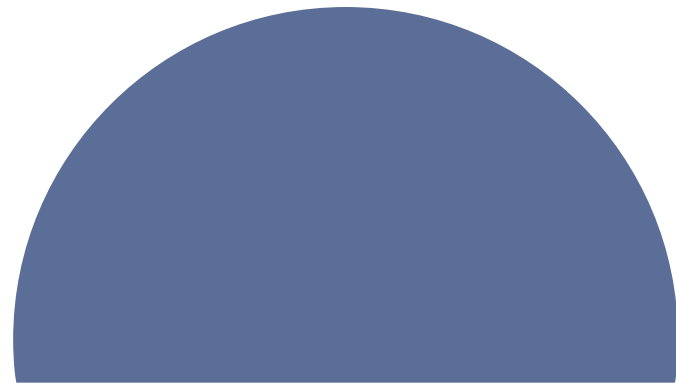


SUPERVISOR
Mohamed Salah



OBJECTIVE

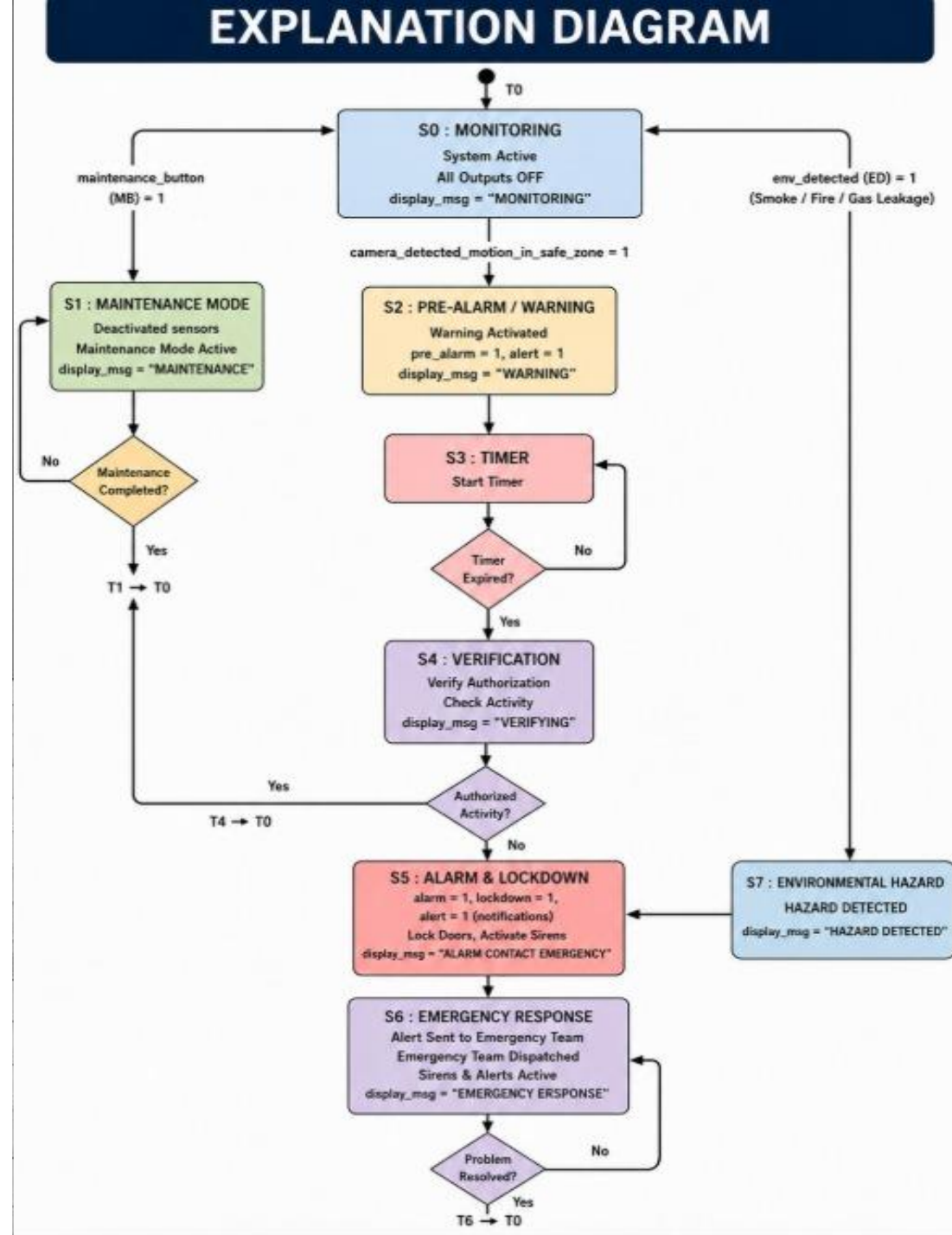
The objective of this project is to design a smart security system for protecting museum using a Finite State Machine (FSM). The system can activate a pre-alarm, alarm, lockdown, emergency mode, or environmental hazard warning depending on the detected event.



DIAGRAM

This diagram shows all system states, decisions, and transitions.

We used it as the design before implementing the Verilog code.



FSM STATES

The system has eight states:

- S0: Monitoring
- S1: Maintenance Mode
- S2: Pre-Alarm
- S3: Timer
- S4: Verification
- S5: Alarm and Lockdown
- S6: Emergency Response
- S7: Environmental Hazard

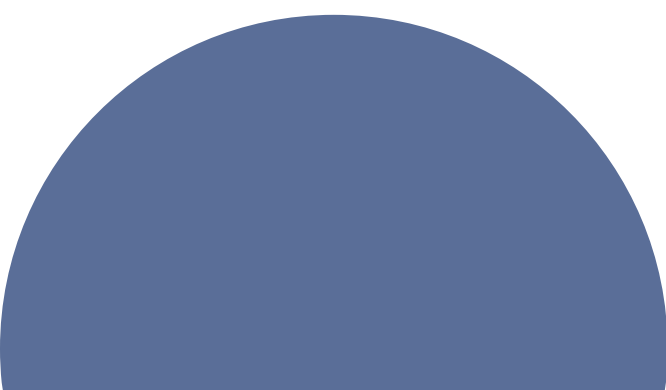
DESCRIPTION

The system starts in the Monitoring state where all outputs are OFF.

If the maintenance button is pressed, the system enters Maintenance Mode.

If camera and motion sensors detect movement inside the safe zone, the system activates a warning, waits for a timer, verifies the person's authorization, and then either returns to monitoring or activates alarm and lockdown.

If fire, or gas leakage is detected, the system immediately enters Alarm & lockdown then Emergency response.



INPUTS AND OUTPUTS

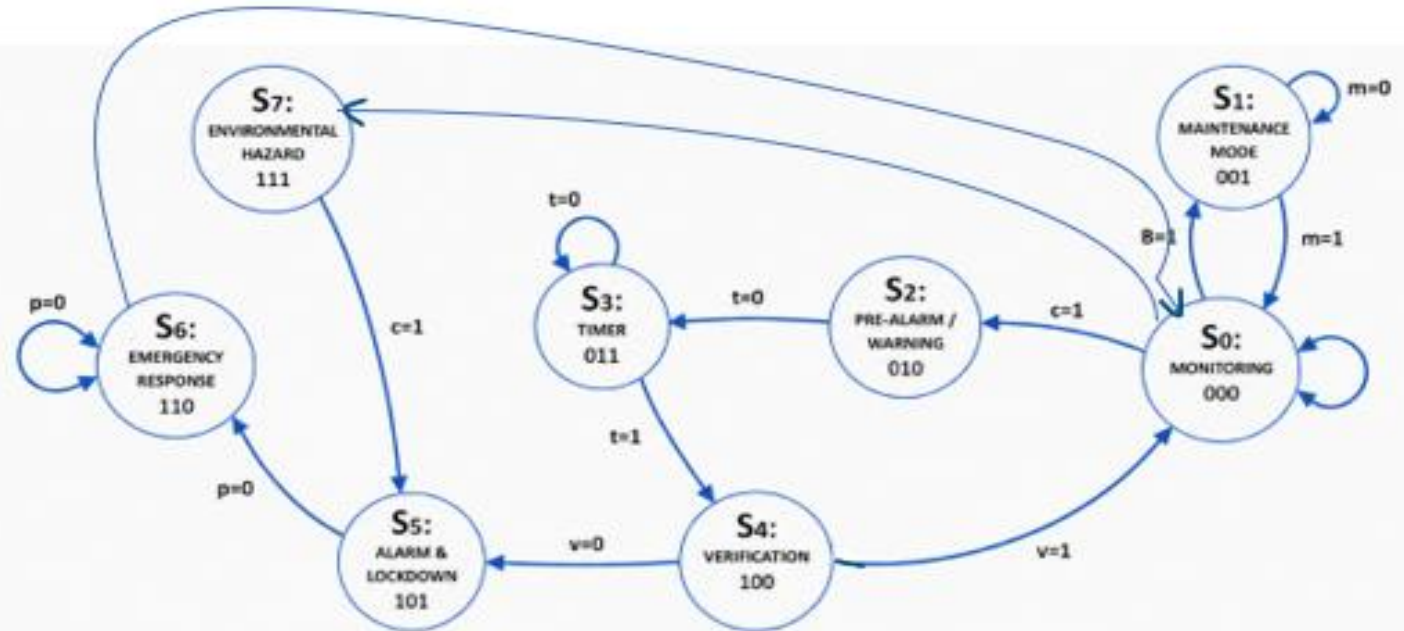
```
module muse(  
    input clk,  
    input rst,  
    input maintenance_button,  
    input maintenance_completed,  
    input camera_detect,  
    input motion_in_safe_zone,  
    input env_detect,  
    input authorized_activity,  
    input problem_resolved,  
  
    output reg timer,  
    output reg alarm,  
    output reg lockdown, //lockdown on artifact not the place  
    output reg emergency,  
    output reg env_hazard,  
    output reg pre_alarm  
);
```

STATE DIAGRAM

This is the State Diagram.

The system starts from Monitoring.

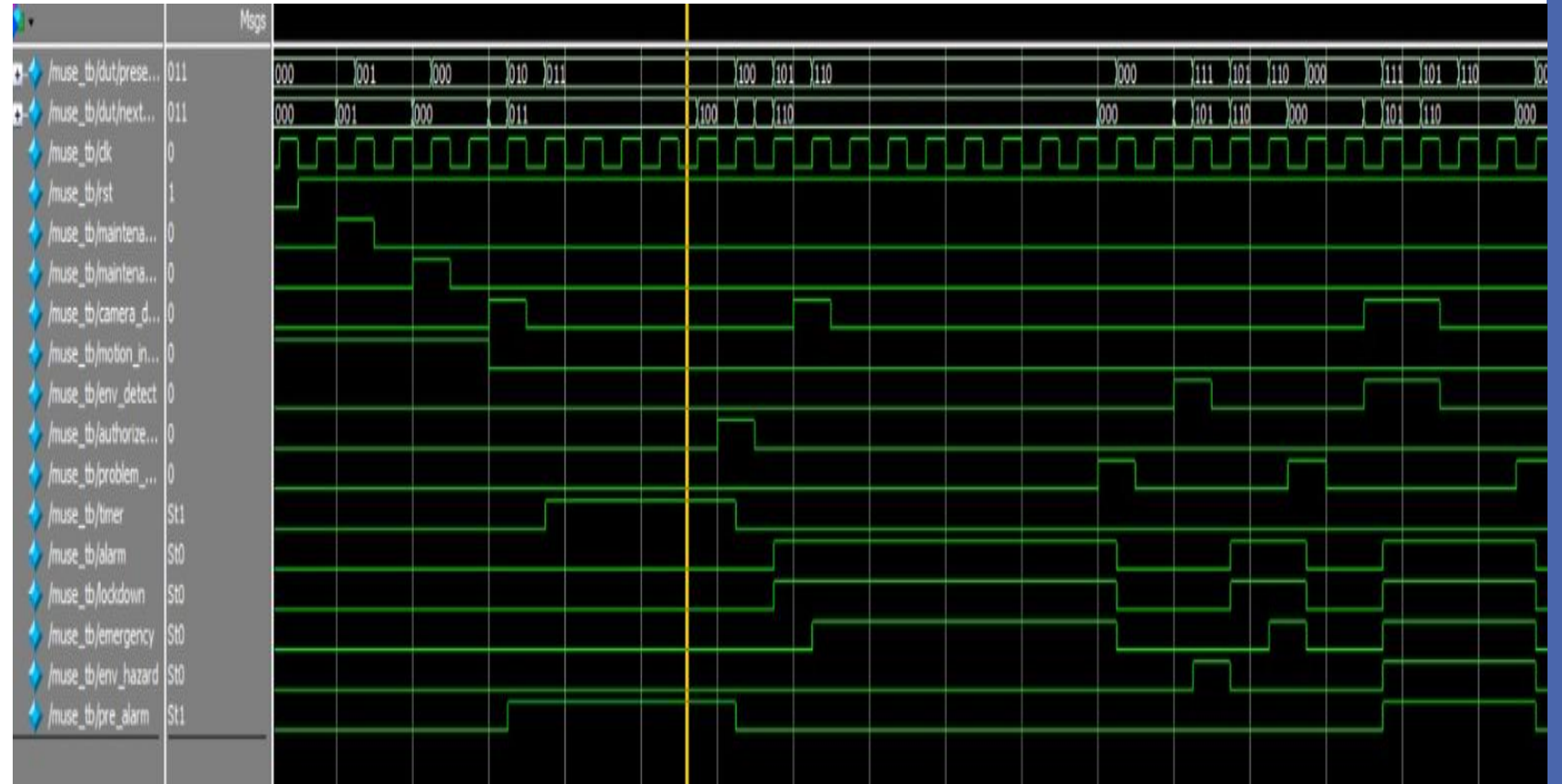
Depending on the input signals, it moves between the different states until it returns to Monitoring again.



Symbol	Meaning
m	m : maintane done
c	c : camera detected
t	t : time expired
v	v : verification done
B	B : maintenance bypass
p	p : problem solved

SIMULATION

- The simulation confirms that the FSM transitions correctly between states.
- The outputs respond as expected for each scenario.
- The timer, alarm, lockdown, emergency, and environmental hazard signals are activated only when their corresponding conditions are satisfied.



CONCLUSION

In conclusion, we successfully designed and simulated a Museum Security System using Verilog and Finite State Machine principles.

The system can handle multiple security scenarios and provides an automatic response to improve the protection of museum artifacts.